

Reinforcement Learning for Quantum Sensing: Phase and Field Estimation Across Quantum Platforms

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Reinforcement learning (RL) has emerged as a powerful paradigm for the optimization of quantum sensors operating in complex, noisy, and time-dependent environments. This article reviews the application of RL to quantum sensing tasks with an emphasis on phase and field estimation. We focus on platforms relevant to contemporary quantum technology, including quantum dots, trapped ions, and neutral atoms. The theoretical framework combines Bayesian inference, Fisher information, and partially observable Markov decision processes (POMDPs), allowing RL agents to adaptively design probe states, control sequences, and measurement strategies. We discuss model-free and model-aware RL approaches, provide pseudocode for representative algorithms, and analyze practical considerations for experimental integration. The results demonstrate how RL can autonomously approach quantum-limited sensitivity and offers a unifying methodology for next-generation quantum metrology.

I. INTRODUCTION

Quantum sensing exploits the coherent dynamics of quantum systems to estimate physical parameters such as phases, frequencies, and fields with extreme precision. In idealized scenarios, the achievable precision is bounded by the quantum Cramér–Rao inequality, which is determined by the quantum Fisher information (QFI) of the probe state. In practice, however, realistic sensors must operate under noise, decoherence, control imperfections, and incomplete information about the parameter of interest.

Adaptive sensing strategies based on Bayesian inference have proven effective in approaching quantum limits, but the design of optimal control and measurement policies quickly becomes intractable as the complexity of the sensor increases. Reinforcement learning provides a natural framework for addressing this challenge by enabling an agent to learn adaptive strategies through interaction with the sensing environment.

This work surveys RL-based approaches to quantum sensing, emphasizing phase and field estimation and highlighting concrete implementations in quantum dots, trapped ions, and neutral atom systems.

II. BAYESIAN UPDATES AND POMDP FORMULATION

Adaptive quantum sensing can be formulated as a partially observable Markov decision process (POMDP). The unknown parameter θ is treated as a hidden state, while the agent maintains a belief distribution $b(\theta)$ updated via Bayes’ rule,

$$b'(\theta) = \frac{P(o|\theta, a) b(\theta)}{\int d\theta' P(o|\theta', a) b(\theta')}, \quad (1)$$

where a denotes a control action and o a measurement outcome.

The reward function is typically chosen to reflect information gain or precision, such as the reduction in posterior variance or entropy. Representing the belief state explicitly converts the POMDP into a fully observable belief-space MDP, enabling the use of standard RL algorithms.

III. FISHER INFORMATION AND QUANTUM LIMITS

For a measurement distribution $p(o|\theta)$, the classical Fisher information is

$$F_C(\theta) = \sum_o \frac{1}{p(o|\theta)} \left(\frac{\partial p(o|\theta)}{\partial \theta} \right)^2. \quad (2)$$

The quantum Fisher information F_Q provides an upper bound on F_C over all possible measurements. For pure states,

$$F_Q = 4 \left(\langle \partial_\theta \psi | \partial_\theta \psi \rangle - |\langle \psi | \partial_\theta \psi \rangle|^2 \right). \quad (3)$$

RL policies can be trained to maximize rewards related to Fisher information, thereby driving the sensing protocol toward quantum-limited performance.

IV. REINFORCEMENT LEARNING METHODS

A. Model-Free Reinforcement Learning

Model-free RL methods, such as Q-learning and policy-gradient algorithms, treat the quantum sensor as a black box. These approaches are flexible and robust but often require a large number of interactions to converge.

B. Model-Aware Reinforcement Learning

Model-aware RL exploits differentiable models of the sensor dynamics, allowing gradients to be propagated through quantum evolution and measurement. This significantly improves data efficiency and enables direct optimization of Fisher-information-based objectives.

C. Offline RL and Domain Randomization

Offline RL leverages pre-collected datasets, while domain randomization improves robustness by training policies across a distribution of noise and model parameters. Both techniques are essential for practical deployment on experimental hardware.

V. QUANTUM SENSING PLATFORMS

A. Quantum Dot Sensors

Quantum dots serve as highly tunable charge and spin sensors. RL has been used to automate device tuning and could be extended to adaptive phase and field estimation by learning optimal gate-voltage and measurement protocols under realistic noise conditions.

B. Trapped Ion Sensors

Trapped ions offer exceptional coherence and control, making them ideal for precision sensing. RL agents can optimize laser pulses, compensation fields, and interrogation times, outperforming hand-designed protocols in the presence of decoherence.

C. Neutral Atom Sensors

Neutral atom platforms, including cold-atom interferometers and magnetometers, benefit from RL-driven discovery of nontraditional interferometric sequences. These approaches have demonstrated enhanced sensitivity within fixed experimental constraints.

VI. PSEUDOCODE EXAMPLES

VII. IMPLEMENTATION CONSIDERATIONS

Key practical challenges include data efficiency, reward design, hardware constraints, and robustness to noise and drift. Hybrid strategies combining simulation pretraining and experimental fine-tuning are likely to be essential for scalable deployment.

Algorithm 1 RL-Based Adaptive Phase Estimation

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1: Initialize belief  $b(\phi)$ , policy  $\pi$ 
2: for each episode do
3:   for  $t = 1$  to  $T$  do
4:     Select action  $a_t = \pi(b(\phi))$ 
5:     Perform measurement, observe  $o_t$ 
6:     Update belief  $b(\phi)$  via Bayes' rule
7:     Compute reward  $r_t$ 
8:     Update policy  $\pi$ 
9:   end for
10: end for
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VIII. CONCLUSION

Reinforcement learning provides a versatile and powerful framework for quantum sensing. By unifying Bayesian inference, Fisher information, and adaptive control, RL enables autonomous discovery of high-performance sensing protocols across a wide range of quantum platforms. As quantum hardware matures, RL-driven sensing is poised to become a central tool in quantum metrology.